

Social Networks and Success in the YouTube Creator Economy

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Abstract

This paper maps the collaboration network of English-language YouTube commentary creators and examines how a creator’s network position relates to their success. The dataset combines YouTube Data API records with 2,917 hand-verified collaborations among 1,391 creators, where a collaboration is a video featuring two or more creators each making a meaningful contribution. This definition is stricter than the subscription, comment, or hyperlink ties used in prior YouTube network studies. The network is sparse and highly disconnected, with a core-periphery-like structure and slight homophily by race and gender. Degree and weighted-degree centrality are positively associated with view and subscriber counts; betweenness and eigenvector centrality show weaker associations. Channel age and total video count are also associated with success, with video count showing the stronger association. Many high-success creators sit on the network periphery without observed collaborations, suggesting that collaboration is one path to success among several. Because the analysis is observational and cross-sectional, the results are descriptive associations rather than causal effects.

1 Introduction

YouTube has become a major venue for independent media work, with creators monetizing through ad revenue, channel memberships, merchandise, and sponsorships (Peres et al., 2023; Radionova and Trots, 2021; Budzinski and Gaenssle, 2018). Many of these creators collaborate, appearing in one another’s videos and reaching one another’s audiences. This paper asks how a creator’s position in the network of those collaborations relates to their success.

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Prior work on YouTube has typically used subscription links, comments, or hyperlinks to define ties between users (Mislove et al., 2007; Rieder et al., 2023). These ties are convenient because they can be scraped at scale, but they are noisy proxies for direct creator-to-creator interaction: a subscription does not require any contact between subscriber and creator, and a hyperlink does not require either party to know the other. Two prior studies are closer to ours in focusing on subsets of YouTube. Mogallapu (2011) studied a smaller community of video bloggers, including both subscriber and personal-communication ties. Koch et al. (2018) used face-recognition software to find videos in which two creators co-occur, a definition that also captures pre-recorded clips of one creator played in another’s video. We use a stricter definition: a collaboration is a video in which two or more creators each make a meaningful contribution, verified by hand. Hand-verification is costly, but it removes the noisy cases that automated proxies cannot.

The setting is the English-language YouTube commentary genre, where individual creators predominate and collaborations are a common feature of content production. The sample contains 1,391 creators and 2,917 hand-verified collaborations. The network is sparse and highly disconnected, with a core-periphery-like structure and slight homophily by race and gender, consistent with prior findings on YouTube network structure (Mislove et al., 2007; Warmbrodt et al., 2008).

We find that degree and weighted-degree centrality are positively associated with views and subscribers, while betweenness and eigenvector centrality show weaker associations. The associations between network position and success are real but modest. Many highly successful creators sit on the periphery without observed collaborations at all, suggesting that collaboration is one path to success among several rather than a necessary condition.

Because the analysis is observational and cross-sectional, these results should be read as descriptive associations. We cannot rule out reverse causality (successful creators attract more collaborations) or omitted-variable bias (creator quality, niche, production resources). The contribution of this paper is the hand-curated network and the descriptive map it makes possible.

2 Related Literature

Two strands of literature inform this paper. The first is the body of work in economics on how an agent’s position in a social or economic network shapes their outcomes, including labor markets, technology diffusion, and inequality (Jackson et al., 2017). From this literature we draw two recurring themes that organize our analysis. Homophily, the tendency of similar agents to connect with one another, has

been documented across many settings and is a feature of both offline and online networks (Tarbush and Teytelboym, 2012). Centrality measures are the standard tools for characterizing an agent’s structural position; we use degree (the number of direct connections), weighted degree (degree weighted by connection intensity), betweenness (the extent to which an agent lies on shortest paths between other agents), and eigenvector centrality (which weights connections by the connectedness of neighbors).

The second strand is the empirical study of online social networks. Across platforms, several findings recur. Online networks tend to have heavy-tailed degree distributions in which a small number of nodes have many connections while most have few. They tend to have core-periphery-like structure, with a densely connected core surrounded by a larger periphery of weakly connected or isolated nodes. And they tend to display moderate homophily by demographic, geographic, or topical attributes. These features appear robustly across Twitter (Himmelboim et al., 2017), Twitch (Dux, 2018), and YouTube (Mislove et al., 2007), despite very different platform affordances and tie definitions. The English-language YouTube commentary collaboration network we examine below shares these features.

3 Data Collection and Description

3.1 YouTube Data API and Commentary Creator Sample

The data for this study were collected through a combination of the YouTube Data API and manually checking YouTube pages, focusing on the commentary creator YouTube social network. Data were collected for both the links (collaborations) and nodes (creators) in the network.

The dataset contains two kinds of data:

1. Link Data: collaboration link (hosting channel, guest channel), collaboration date, collaboration views, collaborator is commentary creator boolean
2. Node Characteristics Data: channel ID, title, description, publishing date, country, total view count, subscriber count, number of videos, topic categories, sub-genre, video types, gender, race, latest video publish date, made for kids status, sub count is hidden boolean

Subscriber counts are reported by the API for all creators in our sample regardless of whether the count is publicly visible on YouTube; we use the reported counts directly.

A commentary creator is defined as an individual (or small group) who produces and shares videos on the YouTube platform, focusing on providing commentary, analysis, or opinions on a variety of subjects. This definition was used to manually classify

creators as either commentary (creators having five or more commentary videos on their channel) or non-commentary creators and restrict our sample to those who are commentary creators. Further, some practical restrictions were included, such as a minimum of 5 video uploads, 1000 subscribers, and only English language creators. The minimum of 1,000 subscribers was chosen because YouTube Partner Program eligibility requires 1,000 subscribers and 4,000 watch hours (Perelli, 2023).

Further, because we are interested in individual, independent creators, we excluded organization-run channels (e.g., news channels) from our sample. A collaboration is defined as a video created by two or more distinct creators who all make meaningful contributions to the video.

We collapse the collaboration videos into a weighted directed graph. Each node is a creator. Each directed edge is a unique ordered host-guest pair, running from the host channel (the channel that uploaded the video) to the guest creator (the creator visiting that channel). The edge weight is the number of qualifying collaboration videos uploaded by that host featuring that guest before the data collection date. Channel-level statistics (subscriber and view counts, video count, channel age) are the values reported by the YouTube Data API at the time of collection in August 2023.

3.2 Sample Construction and Coding

The starting list of commentary creators was the Fandom YouTube commentary creator list (a community-edited wiki accessed in August 2023), which at the time of collection contained 2,462 creators. We manually checked each entry against the commentary creator definition above (five or more commentary videos), the inclusion criteria (English-language, at least five video uploads, at least 1,000 subscribers), and the exclusion of organization-run channels. This produced a final node list of 1,391 individual commentary creators, with a combined 1.1 million uploaded videos as recorded in the YouTube Data API.

Collaboration links were found in two stages. First, an algorithmic title-keyword filter (searching for terms such as “w/”, “ft.”, “feat.”, “featuring”, “collab”, “along-side”, and “@”) applied to the 1.1 million video titles produced approximately 100,000 candidate collaboration videos. Second, we manually reviewed each candidate using the video title, description, video page, and the video content itself where necessary, retaining only those in which two or more distinct creators each made a meaningful contribution. Videos in which a guest creator was only mentioned, briefly shown via pre-recorded clip, or otherwise not personally participating were excluded. The verified set contains 2,917 collaboration videos.

We coded gender, race, sub-genre, and video type manually, based on visible cues

in each channel’s videos, channel description, and About page. Two coders (the author and one helper) applied shared informal criteria; no formal coding rubric was written. Where cues were ambiguous or absent, the value was coded as “unknown” rather than guessed. Two limitations apply: externally coded demographic categories may not match creator self-identification, and intercoder reliability was not formally measured (ambiguous cases were resolved by discussion). The “unknown” categories are reported separately from substantive categories throughout. These variables should therefore be interpreted as externally coded public presentation categories, not as verified self-identifications.

3.3 Variables and Data Description

Tables 1, 2, and 3 show summary statistics for the nodes and links in the network.

Table 1: Categorical Summary Node Statistics

	New ID	Channel Country	Gender	Race
count	1391	1193	1391	1391
unique	1391	40	5	6
top		US	male	white
freq		803	1144	686

Table 2: Numerical Summary Node Statistics

	View Count	Subscriber Count	Video Count
count	1391	1391	1391
mean	209691279	789408	727
std	976182445	3314647	1654
min	8301	1060	5
25%	5518970	49100	101
50%	32590743	235000	267
75%	128854776	675000	704
max	29097272573	111000000	32076

Table 3: Summary Link Statistics

	Host ID	Guest ID	Collab Video Date	Collab Video Views
count	2917	2917	2917	2917
unique	425	498	2636	
top	861	779	2022-07-18T09:01:31Z	
freq	134	100	5	
mean				340551
std				1295848
min				60
25%				14296
50%				59981
75%				206801
max				22577012

Figures 1 to 5 show the distributions for a few categorical variables. For race, gender, and country, the top categories are white, male, and US by a large margin. With the topic categories variable, it should be noted that each node has multiple topic categories assigned to it. In topic categories, the three largest are Entertainment, Video Game Culture, and Lifestyle.

Figure 1: Distribution of Gender

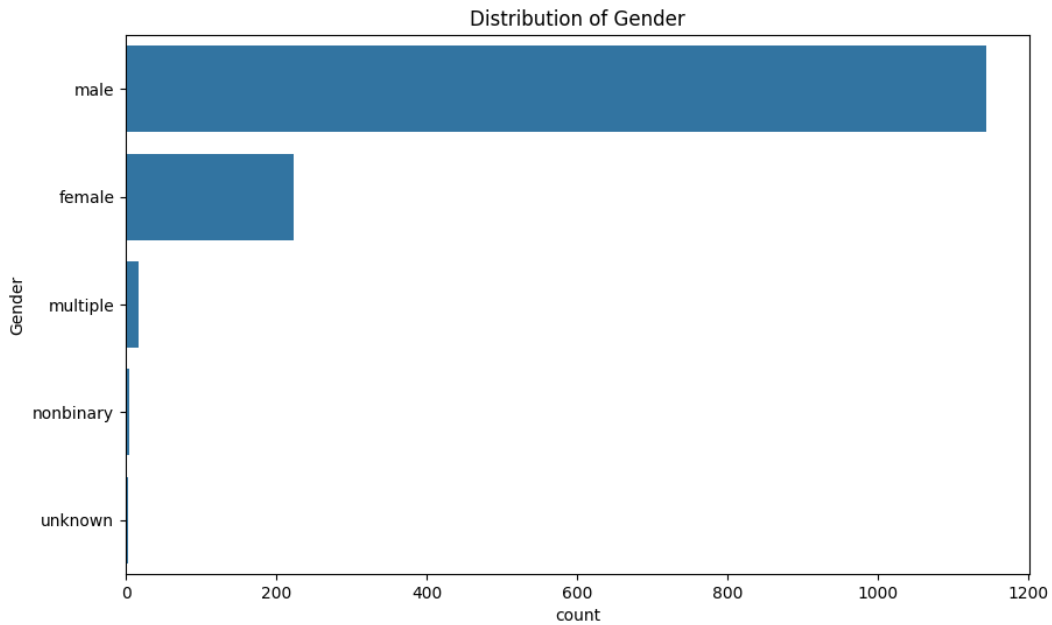


Figure 2: Distribution of Race

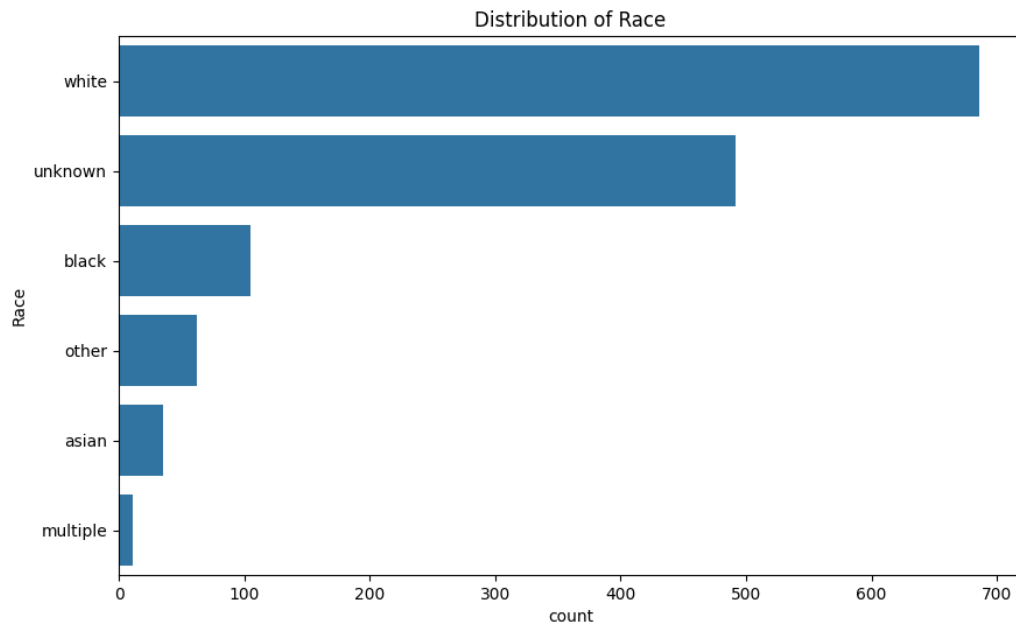


Figure 3: Distribution of Topic Categories

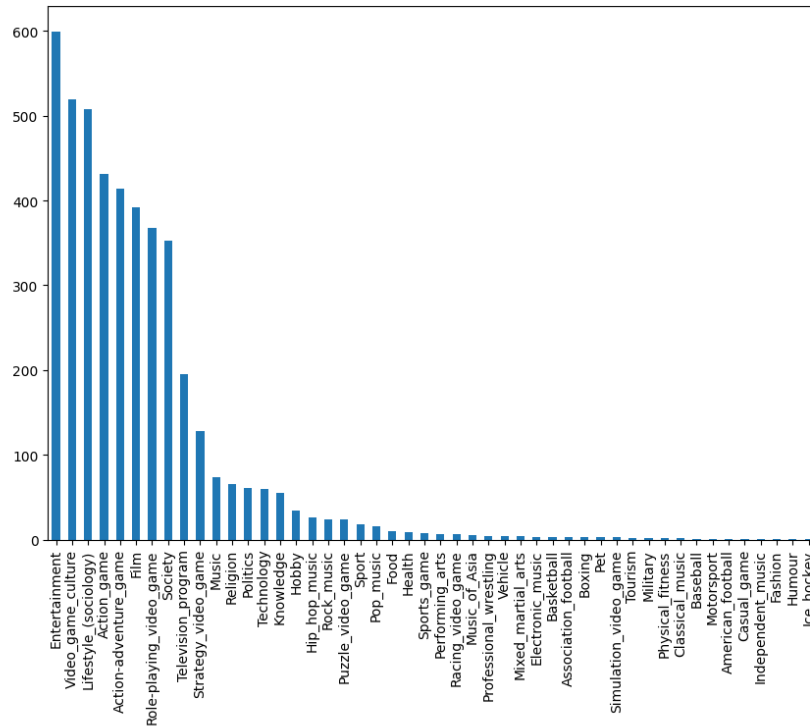


Figure 4: Distribution of Countries

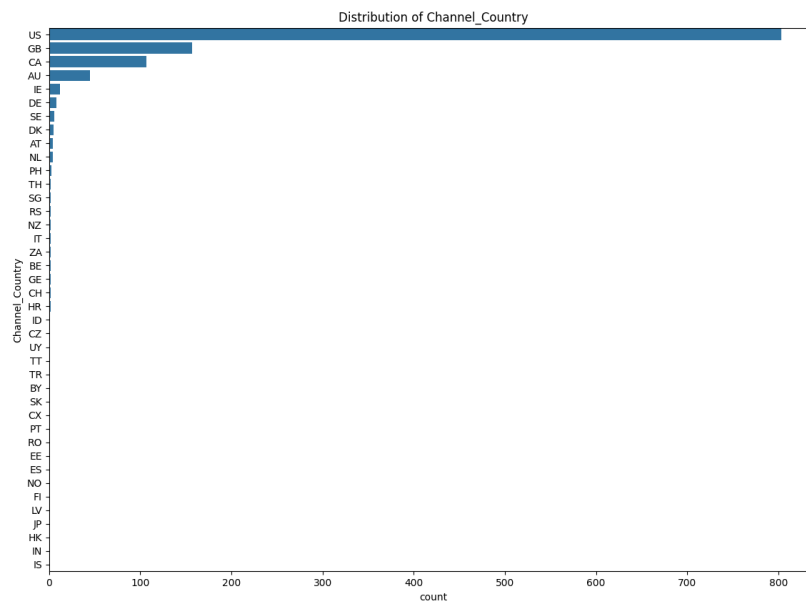
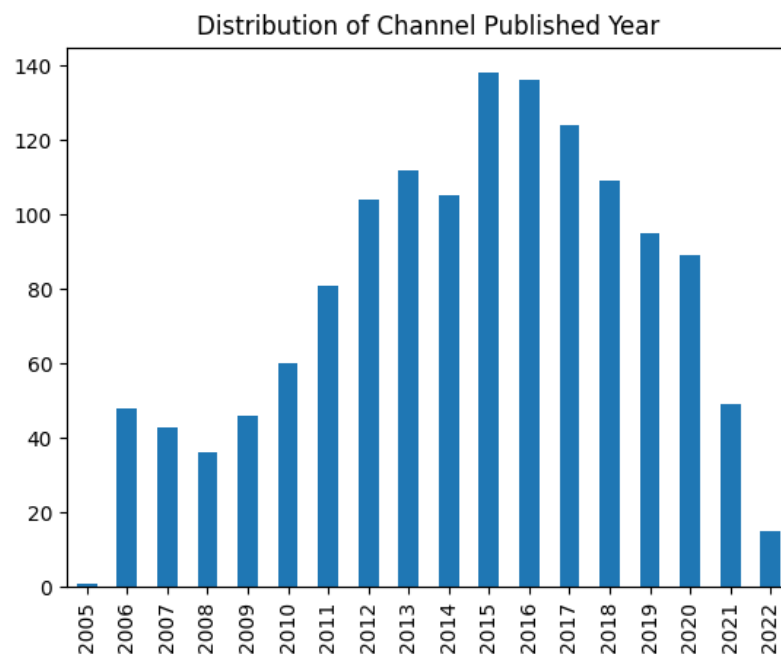


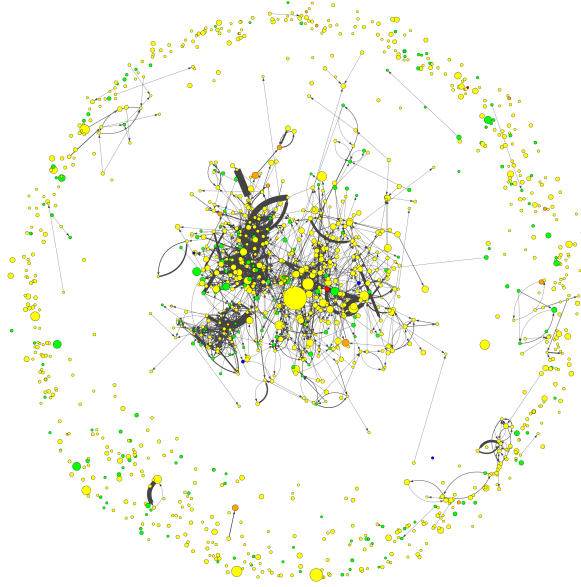
Figure 5: Distribution of Channel Published Year



4 Results

4.1 Network Description

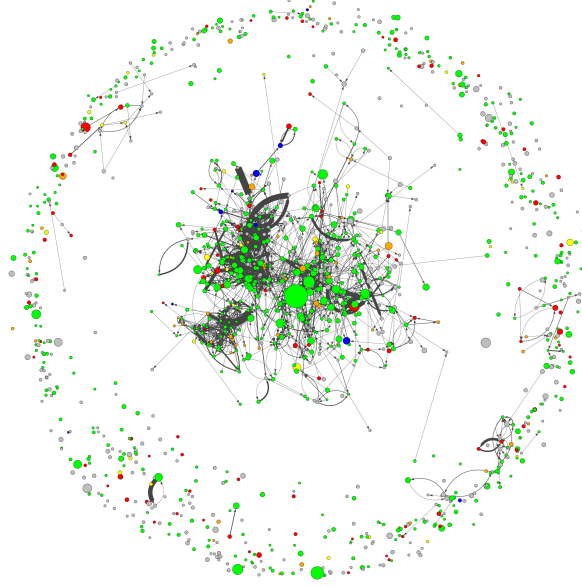
Figure 6: Graph of Full Network: Node Size by Success, Node Color by Gender



Key: 'male' - yellow, 'female' - green, 'nonbinary' - blue, 'unknown' - red, 'multiple' - orange

Figures 6 and 7 show the directed collaboration network. Node size is a success proxy constructed by min-max normalizing each creator's subscriber count and view count to $[0, 1]$ and summing the two. Node color encodes gender (Figure 6) or race (Figure 7), and edge thickness encodes the number of collaboration videos between the two creators. The sample is majority male and white. The network is sparsely connected with many isolated nodes, roughly fitting a core-periphery-like structure consistent with prior findings on YouTube (Mislove et al., 2007; Warmbrodt et al., 2008) though not with Mogallapu (2011). While success is concentrated near the center of the network, many high-success creators also appear on the periphery without observed collaborations.

Figure 7: Graph of Full Network: Node Size by Success, Node Color by Race



Key: 'white' - green, 'black' - red, 'asian' - yellow, 'other' - orange, 'multiple' - blue, 'unknown' - grey

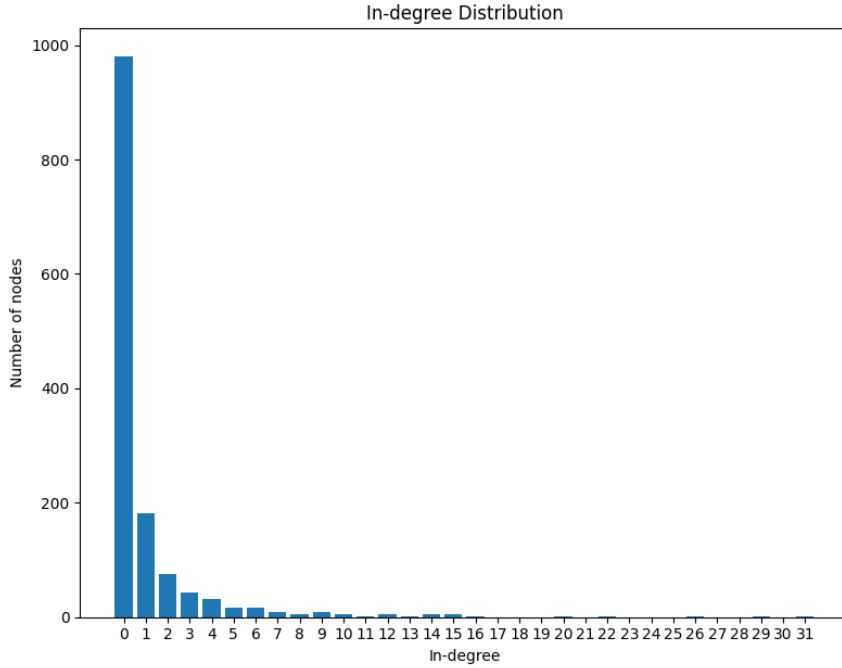
Table 4: Summary Network Statistics

Metric	Value
Number of vertices	1391
Number of edges	1339
Density	0.001
Average degree	1.925
Average indegree	0.963
Average outdegree	0.963
Number of components	1198
Number of non-isolated components	32
Average path length	6.713
Diameter	21
Average clustering coefficient	0.272

Table 4 reports summary statistics for the full network. The network is sparse (density 0.001) and highly disconnected. Per Table 3, 425 creators appear as a host and 498 as a guest in at least one observed collaboration, so most of the 1,391 creators in the sample have few or no observed collaborations. The reported number of components (1,198) is the count of strongly connected components and overstates the count of fully isolated creators, since a creator with collaborations can still form a singleton strong component when no directed cycle runs through them. The reported average path length, diameter, and clustering coefficient are computed over connected

pairs only and do not characterize the disconnected structure as a whole. Within the connected portion of the network, reachable creators are separated by relatively few steps on average. Many creators without observed collaborations are nonetheless fairly successful.

Figure 8: Histogram of In-Degree



Figures 8 and 9 show the in-degree and out-degree distributions. Most creators have neither uploaded a collaboration video (out-degree zero) nor been featured as a guest by another creator (in-degree zero). Figure 10 plots the log-log complementary cumulative distribution function for both, where the y-axis gives the fraction of creators with degree above x .

Table 5: Assortativity Measures

Metric	Value
Degree Assortativity	0.12
Gender Assortativity	0.14
Race Assortativity	0.20
Country Assortativity	0.09
Year Created Assortativity	0.04

Table 5 reports degree, gender, race, country, and year-of-channel-creation assortativity. The degree assortativity is 0.12, indicating only a slight preference for creators

Figure 9: Histogram of Out-Degree

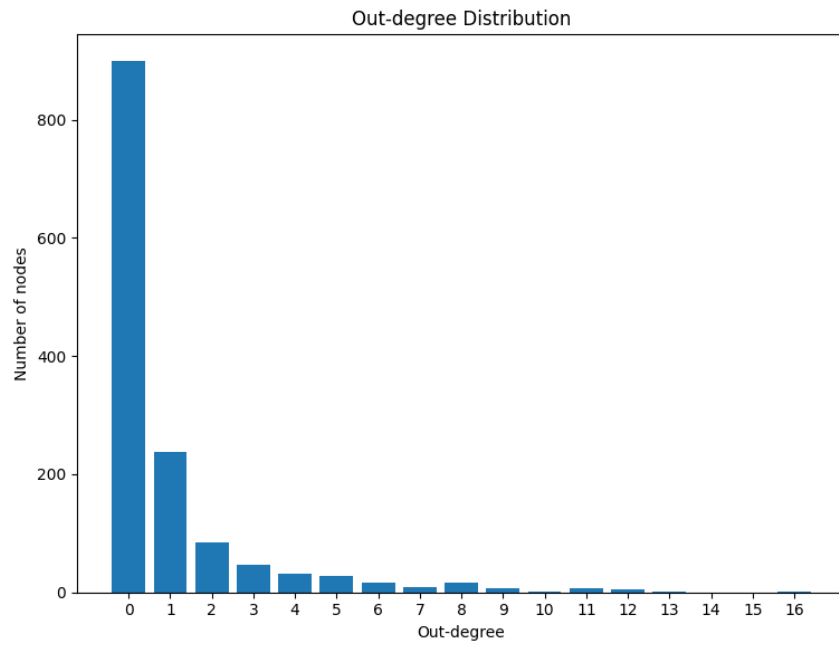
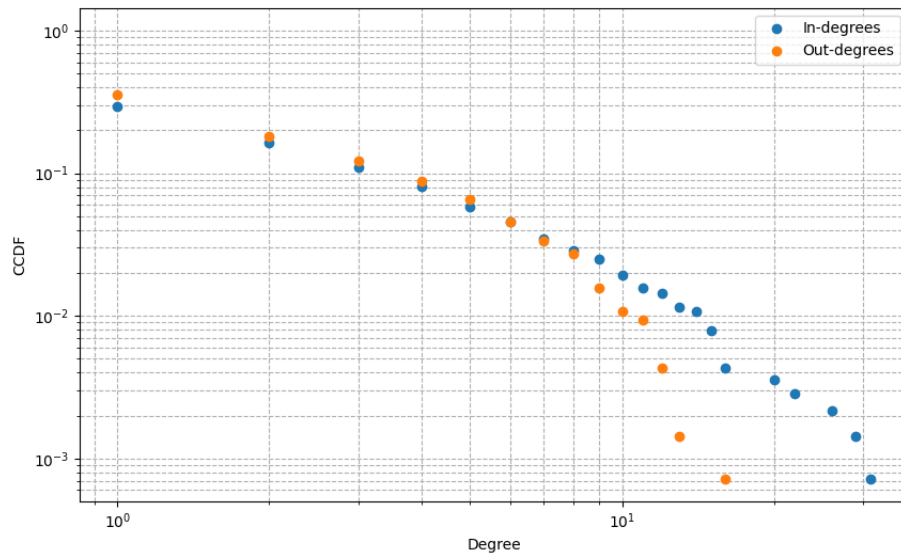


Figure 10: Degree Distribution: log-log CCDF vs. degrees



to collaborate with others of similar degree. Race assortativity is the highest at 0.20, followed by gender at 0.14. These are raw assortativity values; in a sparse network where most creators are male and white, raw values may overstate within-group preferences relative to a degree-preserving null model.

Figure 11 shows sub-networks for the five largest topic categories. Node size encodes success and color encodes gender. The networks for each topic category resemble the overall network: sparsely connected with a small number of larger components and many isolates.

4.2 Network Analysis

Table 6: Correlations between Centrality and Success Measures

	Subscriber Counts	View Counts	Success
Degree	0.271	0.249	0.263
Weighted Degree	0.274	0.259	0.270
Betweenness	0.196	0.198	0.198
Eigenvector	0.096	0.107	0.101

Table 6 reports Spearman correlations between centrality measures and the success proxy. Degree centrality has a correlation of 0.263 with success, and weighted degree, which counts repeated collaborations within a host-guest pair, has a correlation of 0.270; the small difference between the two suggests that the data do not strongly distinguish “more unique partners” from “more videos with the same partner.” Betweenness centrality has a correlation of 0.198, lower than degree, consistent with creators who bridge across components or sub-genres having some additional access to audiences but not dominating the picture. Eigenvector centrality is the weakest at 0.101, which suggests that proximity to highly connected creators is not on its own predictive of success. All four associations are positive but modest in magnitude, and many high-success creators sit far from the network core.

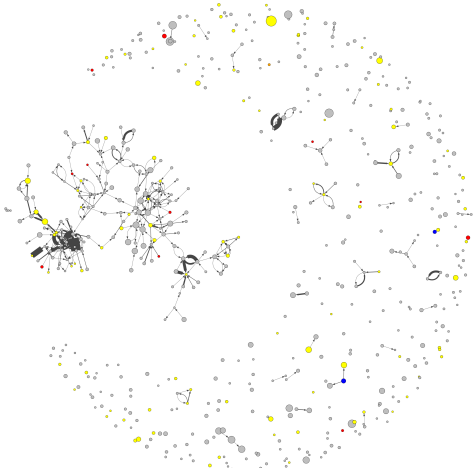
Figure 12 shows the graphs with node size based on success and node color based on the different centrality measures (lighter is higher centrality, darker is lower).

4.3 Individual Analysis

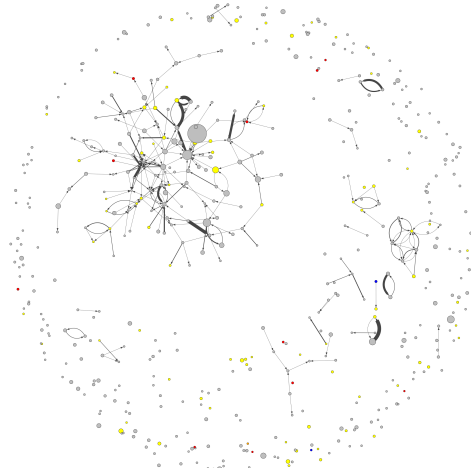
We now examine how individual creator characteristics relate to the success proxy. We consider three categorical variables (gender, race, country) and two continuous variables (total video count and the date the channel was created).

Figure 11: Graphs of Sub-Networks

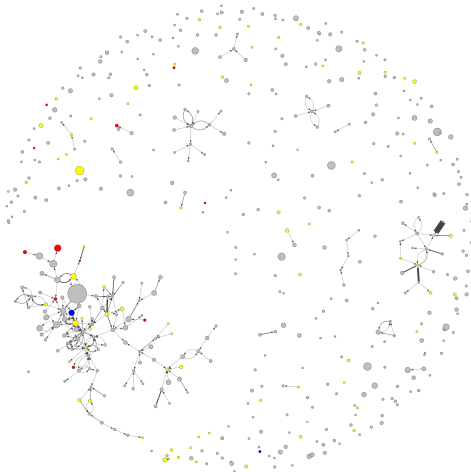
(a) Entertainment Sub Network



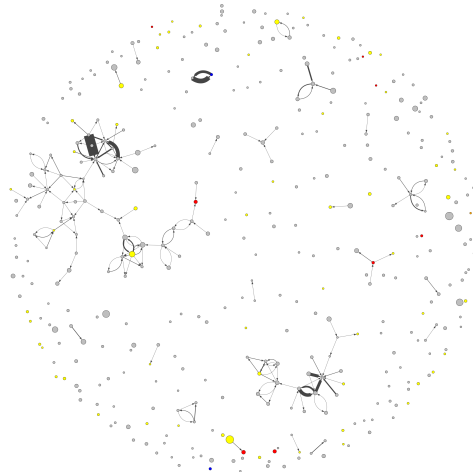
(b) Video Game Culture Sub Network



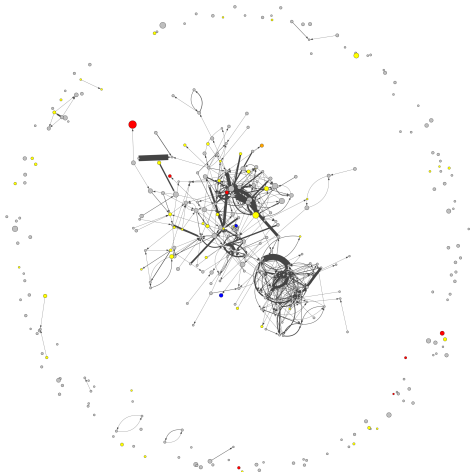
(c) Lifestyle Sub Network



(d) Film Sub Network



(e) Society Sub Network

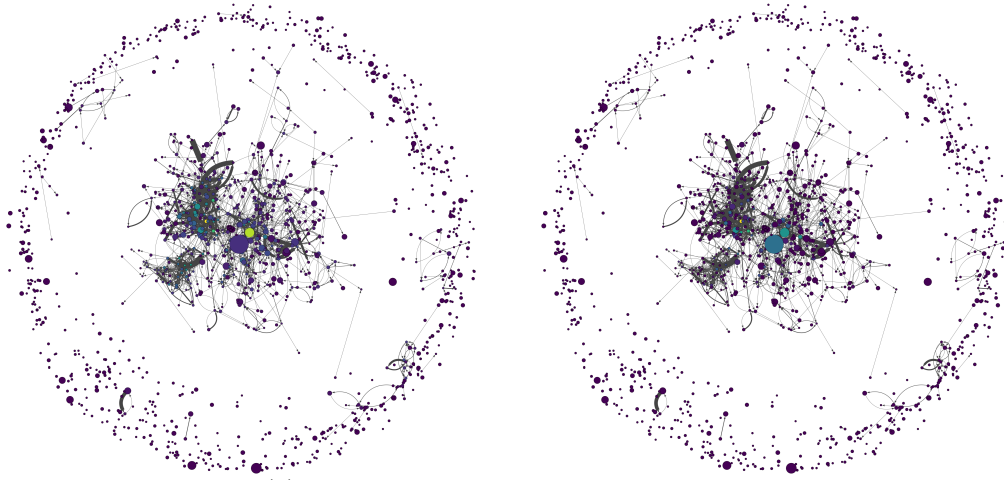


Key: 'male' - grey, 'female' - yellow, 'nonbinary' - blue, 'multiple' - red, 'unknown' - orange

Figure 12: Graphs of Network with Success and Centrality Measures

(a) Success and Degree Centrality

(b) Success and Betweenness Centrality



(c) Success and Eigenvector Centrality

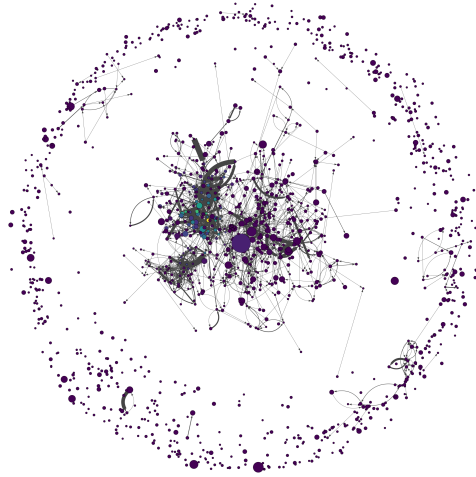


Figure 13: Boxplot of Success by Gender

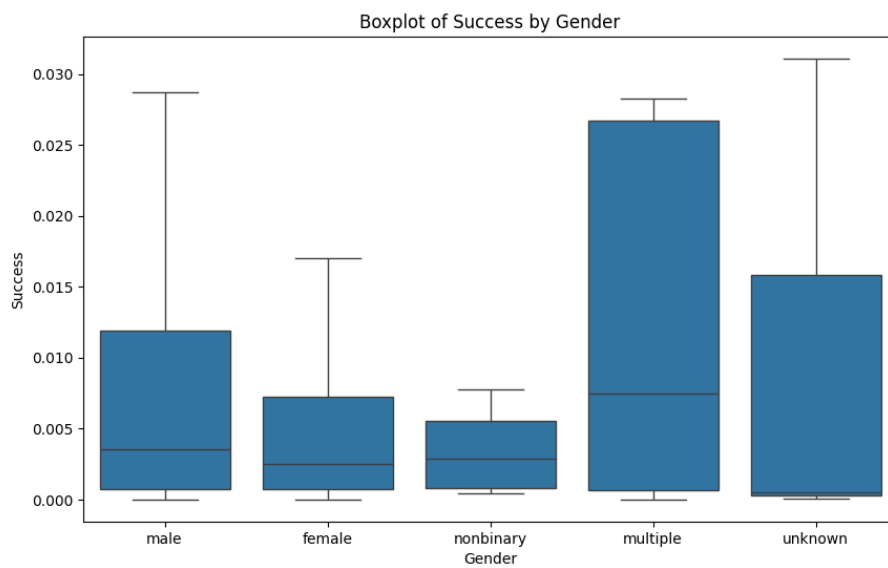


Figure 14: Boxplot of Success by Race

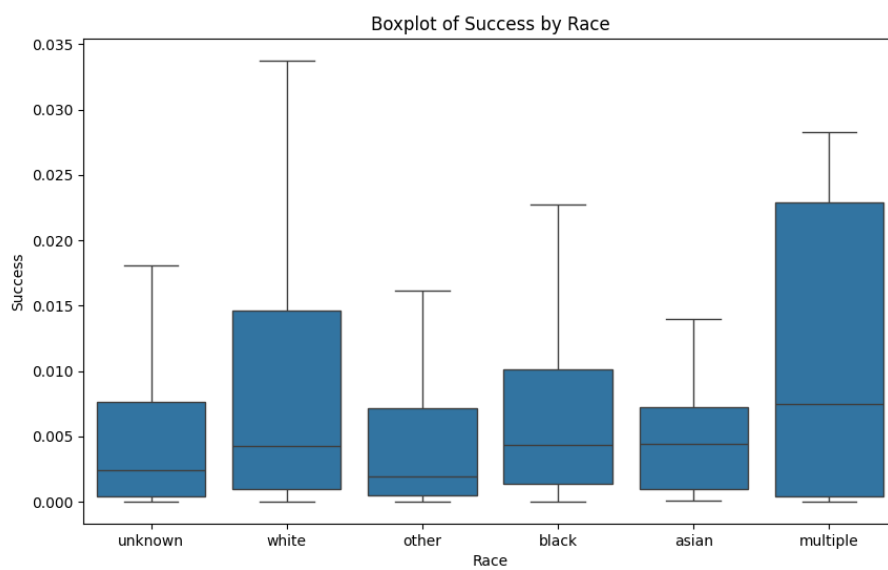
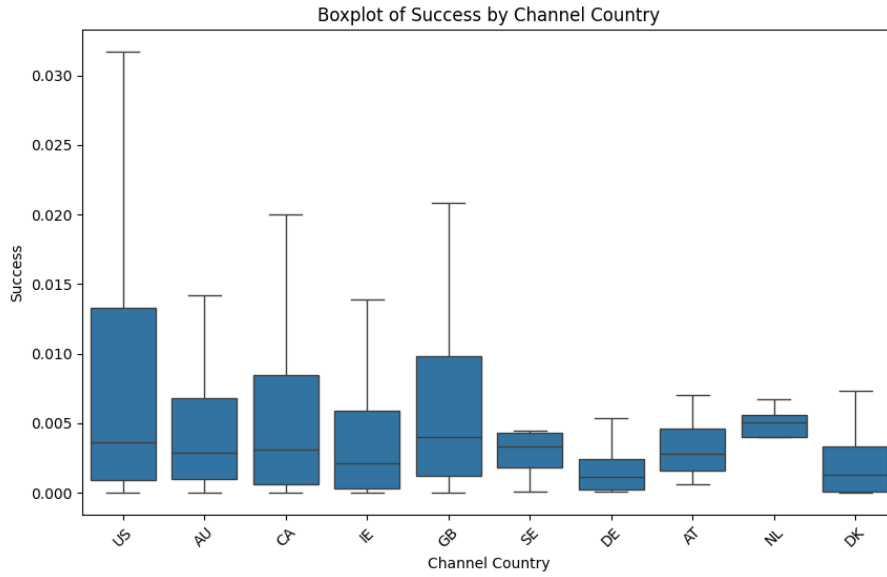


Figure 15: Boxplot of Success by Country



Figures 13, 14, and 15 show success by gender, race, and country, respectively. A Kruskal-Wallis test yields $p = 0.17$ for gender (test statistic 6.32), $p < 0.001$ for race (test statistic 32.57), and $p = 0.28$ for country (test statistic 43.58).

Table 7: Post-hoc pairwise comparison using the Mann-Whitney U test with a Bonferroni correction

	Group1	Group2	P-Value	P-Value Adjusted
0	unknown	white	0.000	0.000
1	unknown	other	0.928	1.000
2	unknown	black	0.003	0.044
3	unknown	asian	0.176	1.000
4	unknown	multiple	0.406	1.000
5	white	other	0.029	0.436
6	white	black	0.815	1.000
7	white	asian	0.544	1.000
8	white	multiple	0.954	1.000
9	other	black	0.061	0.916
10	other	asian	0.316	1.000
11	other	multiple	0.553	1.000
12	black	asian	0.603	1.000
13	black	multiple	0.821	1.000
14	asian	multiple	0.699	1.000

Table 7 reports post-hoc pairwise Mann-Whitney U tests for race with Bonferroni correction. The differences involving the “unknown” race category and the “white” and “black” categories are statistically significant after correction. One plausible interpretation is that creators without a coded race may include channels with no consistent on-camera presence (animated channels, group channels, channels with off-camera hosts), since the coders relied on visible cues to assign race. We do not have a separately coded on-camera presence variable, however, so this interpretation conflates two things: whether the creator appears on camera, and whether the coders had enough visible cues to assign a race category.

Gender differences are not statistically significant ($p = 0.17$). Country differences are also not statistically significant ($p = 0.28$), but this test is uninformative given that 803 of the 1,193 creators with a coded country are based in the United States, leaving most non-US country categories with very few creators each. We do not interpret the country result as evidence that success transcends geography; the data does not support comparison across country groups at this granularity.

Table 8: Correlations between Video Count, Channel Published At, and Success Measures

	Channel Published At	Video Count
Success	-0.234	0.406
View Count	-0.260	0.515
Subscriber Count	-0.210	0.320

Table 8 reports Spearman correlations between two creator-level continuous variables and the success measures. The date the channel was created is negatively correlated with success (correlation -0.234), reflecting the expected pattern that older channels (earlier creation dates) tend to have higher cumulative views and subscriber counts. Total video count has a stronger positive correlation with success at 0.406. The two together are consistent with sustained posting volume being more strongly associated with success than channel tenure alone.

5 Conclusion

This paper has mapped the collaboration network of 1,391 English-language YouTube commentary creators using 2,917 hand-verified collaborations and examined how creators’ positions in the network relate to their success. The network is sparse and highly disconnected, with a core-periphery-like structure and slight homophily by race and

gender. Degree and weighted-degree centrality are positively associated with views and subscribers, with correlations near 0.27. Betweenness and eigenvector centrality show weaker associations. Channel age and total video count are also associated with success, with video count showing the stronger correlation. Many high-success creators sit on the network periphery without observed collaborations, suggesting that collaboration is one path to success among several rather than a necessary condition.

Several limitations apply. First, the analysis is observational and cross-sectional. We cannot rule out reverse causality, since successful creators are likely more attractive collaboration partners, nor can we rule out omitted-variable bias from creator quality, niche, or production resources. Second, the sample was constructed by filtering an external creator list and is not a probability sample of the YouTube commentary genre; creators who never appeared on the seed list are absent from our network. Third, the dataset is a 2023 snapshot, and we cannot speak to how the network or its relationship to success has evolved over time. Fourth, demographic categories were coded by external observers from visible cues, which may not match creator self-identification. Fifth, we exclude organization-run channels and non-English creators, so the findings do not generalize to the broader YouTube ecosystem.

Future work could address some of these limitations. Building network snapshots from collaboration dates would allow a partial dynamic analysis even within the existing data. Adding multivariable regressions with controls for channel age, video count, and topic fixed effects would tighten the descriptive associations reported here. Extending the data to other genres and languages, or developing automated tools that match the strictness of our hand-verified collaboration definition, would broaden the scope but require considerable additional effort.

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